Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

Course Title: Basic Electrical and Electronic Circuit Lab

Course Code: 07131102W

Experiment 08

Design and Performance Analysis of a BJT Class-A Pre-Amplifier Circuit

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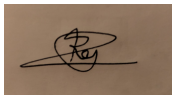
Declaration

I hereby declare that the work presented in this laboratory report entitled **"Design and Performance Analysis of a BJT Class-A Pre-Amplifier Circuit"** is an authentic record of my own work carried out for the course **Basic Electrical and Electronic Circuit Lab (07131102W)**.

This report has been prepared by me independently and has not been submitted elsewhere for the award of any degree, diploma, or credit.

Sylhet, Bangladesh

Date: September 29, 2026



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Objective

- To design and construct a Common-Emitter (CE) amplifier using a voltage-divider biasing network and to measure the stable DC Quiescent Operating Point (Q-Point).
- To inject a small-signal AC sine wave and calculate the practical AC voltage gain (A_v), demonstrating the 180° phase inversion characteristic of the CE topology.
- To analyze the impact of input/output DC blocking (coupling) capacitors and the emitter bypass capacitor on the circuit's performance.
- To sweep the input signal frequency and plot a Bode response curve, identifying the mid-band gain, lower cutoff frequency (f_L), upper cutoff frequency (f_H), and the overall circuit Bandwidth (BW).

Theory and Operational Mechanics

Class-A Amplification

An amplifier is classified as "Class-A" when the active device (BJT) is biased such that output current flows for the entire 360° of the input AC cycle. This requires the Quiescent operating point (Q-point) to be positioned precisely in the center of the active linear region of the transistor's DC load line. While Class-A amplifiers suffer from low theoretical power efficiency, they provide the highest linearity and lowest harmonic distortion, making them the standard choice for low-noise pre-amplification of small signals.

Voltage-Divider Biasing

To prevent thermal runaway and make the Q-point independent of individual transistor current gain variations, a voltage-divider biasing network (R_1 and R_2) is utilized. This establishes a stiff, constant DC potential at the base of the transistor:

$$V_B \approx V_{CC} \left(\frac{R_2}{R_1 + R_2} \right)$$

The DC emitter voltage (V_E) is strictly tied to V_B minus the forward-biased base-emitter barrier voltage ($V_{BE} \approx 0.7\,\text{V}$):

$$V_E = V_B - 0.7\,\text{V} \implies I_E = \frac{V_E}{R_E} \approx I_C$$

The quiescent Collector-Emitter voltage (V_{CE}) is evaluated using Kirchhoff's Voltage Law down the output branch:

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Role of Reactive Components (Capacitors)

- **Coupling Capacitors (C_{in} and C_{out}):** These capacitors act as open circuits to DC, preventing the external source or load from altering the transistor's internal DC bias voltages. However, they act as short circuits to high-frequency AC.
- **Emitter Bypass Capacitor (C_E):** Placed in parallel with R_E, this capacitor provides a low-impedance path to ground for AC signals. Without it, the AC signal would drop across R_E, causing negative feedback (swamping) and reducing voltage gain.

Apparatus and Components

Table 1: List of Equipment and Components Used

Item Description	Specification / Rating	Quantity
Dual Regulated DC Power Supply	0 V to 30 V	1
NPN Transistor	BC547, 2N2222, or equivalent	1
Function Generator	Sine wave generation (used at 1 kHz)	1
Dual-Channel DSO	Digital Storage Oscilloscope	1
Resistors	45.6 kΩ, 9.73 kΩ, 3.22 kΩ, 0.977 kΩ	1 each
Electrolytic Capacitors	10 µF	2
Digital Multimeter (DMM)	Standard	1
Solderless Breadboard	Standard	1

Circuit Diagram

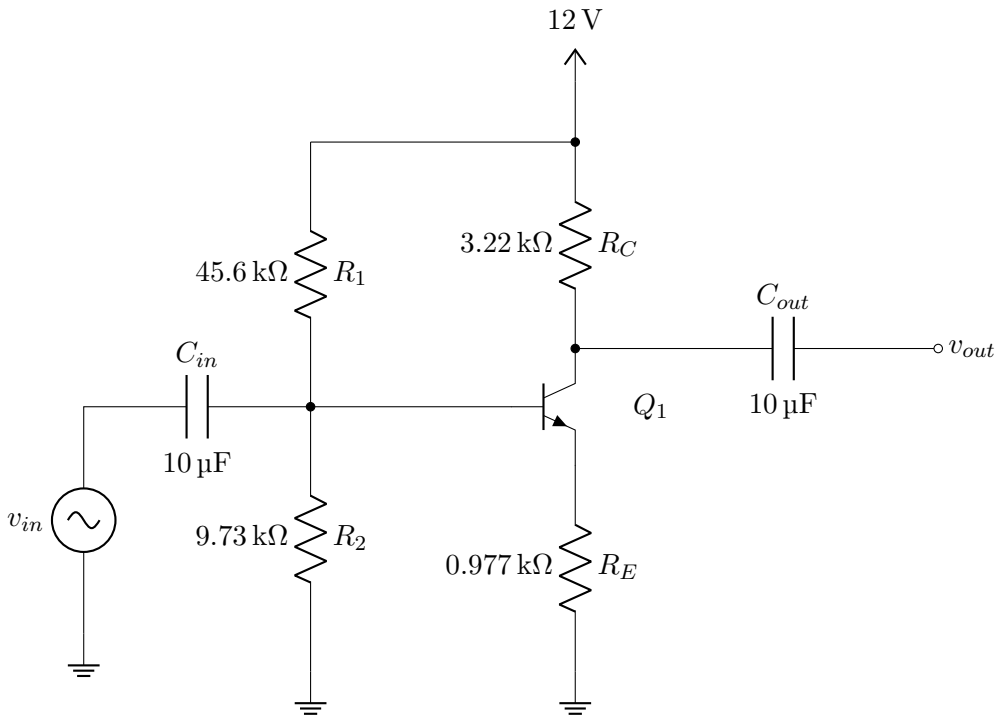


Figure 1: Schematic diagram of the BJT Class-A Common-Emitter Pre-Amplifier utilizing voltage-divider bias.

Experimental Procedure

1. **DC Biasing Configuration:** Construct the circuit according to Figure 1. Initially, keep the Function Generator (v_{in}) disconnected. Energize the circuit by applying $V_{CC} = 12\,\text{V}$ from the DC power supply.

2. **DC Measurement:** Using the DMM in DC Voltage mode, measure the DC voltages at the Base (V_B), Emitter (V_E), and Collector (V_C) relative to ground. Record the values and evaluate

I_C and V_{CE}.

3. **AC Mid-Band Voltage Gain:** Connect the Function Generator to the input coupling capacitor (C_{in}) and apply a 1 kHz sine wave.

4. **Observation:** Connect Channel 1 of the DSO to the input and Channel 2 to the output. Observe the waveform geometries, document the phase shift, and monitor changes across states.

Data Collection

Table 2: DC Quiescent Operating Point (Q-Point) Parameters

Parameter	Theoretical Value	Measured Value	% Error
Base Voltage (V_B)	2.11 V	1.968 V	6.73%
Emitter Voltage (V_E)	1.41 V	1.333 V	5.46%
Collector Voltage (V_C)	7.35 V	7.86 V	6.88%
V_{CE}	5.94 V	6.53 V	9.86%
Collector Current (I_C)	1.44 mA	2.49 mA	72.92%

Oscilloscope Waveforms

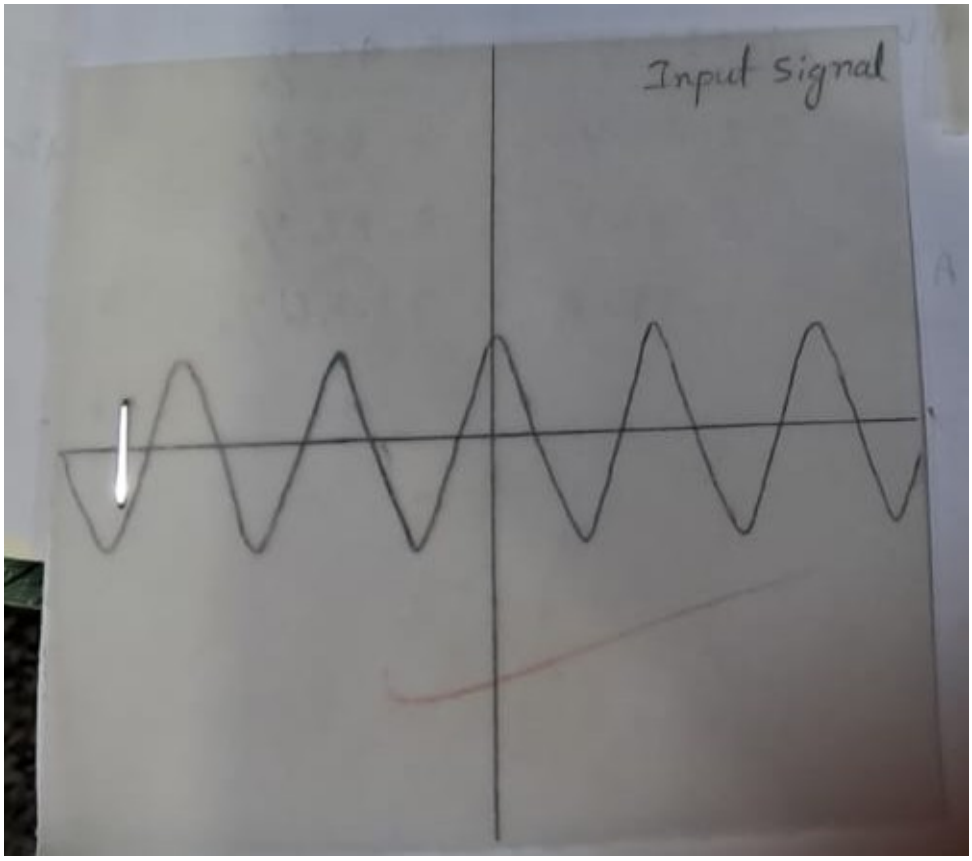


Figure 2: Oscilloscope trace record showing the input signal waveform.

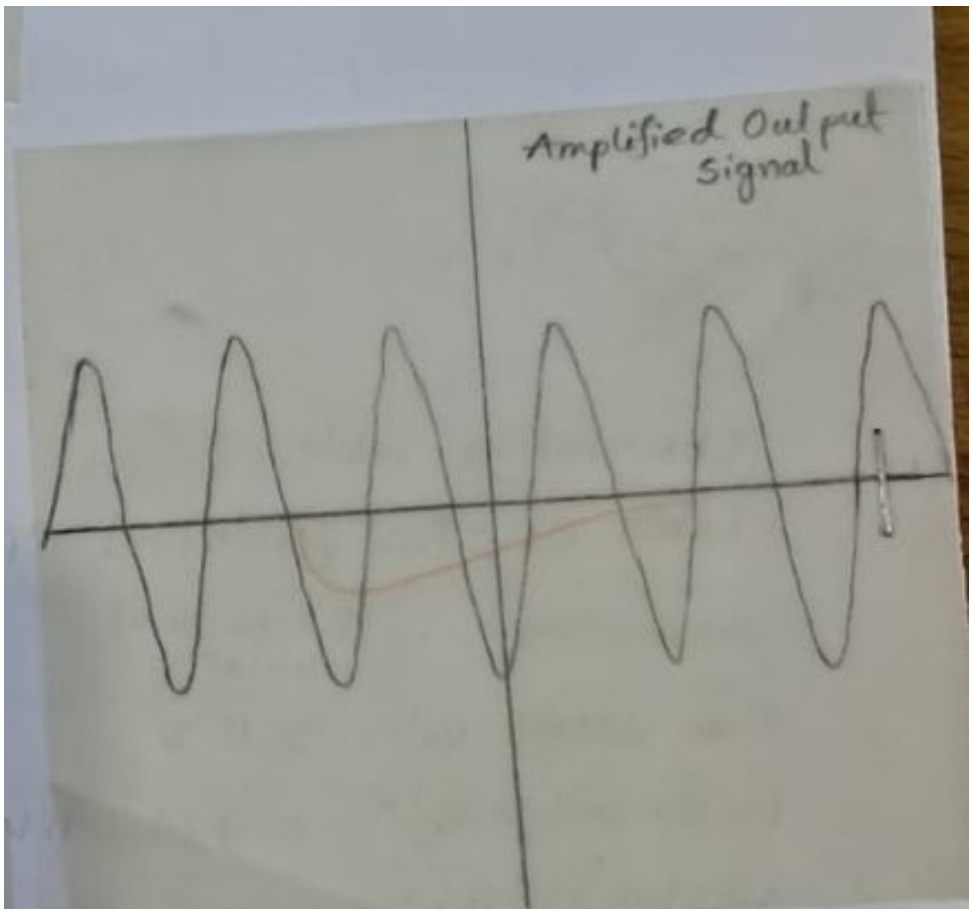


Figure 3: Oscilloscope trace record showing the amplified output signal waveform.

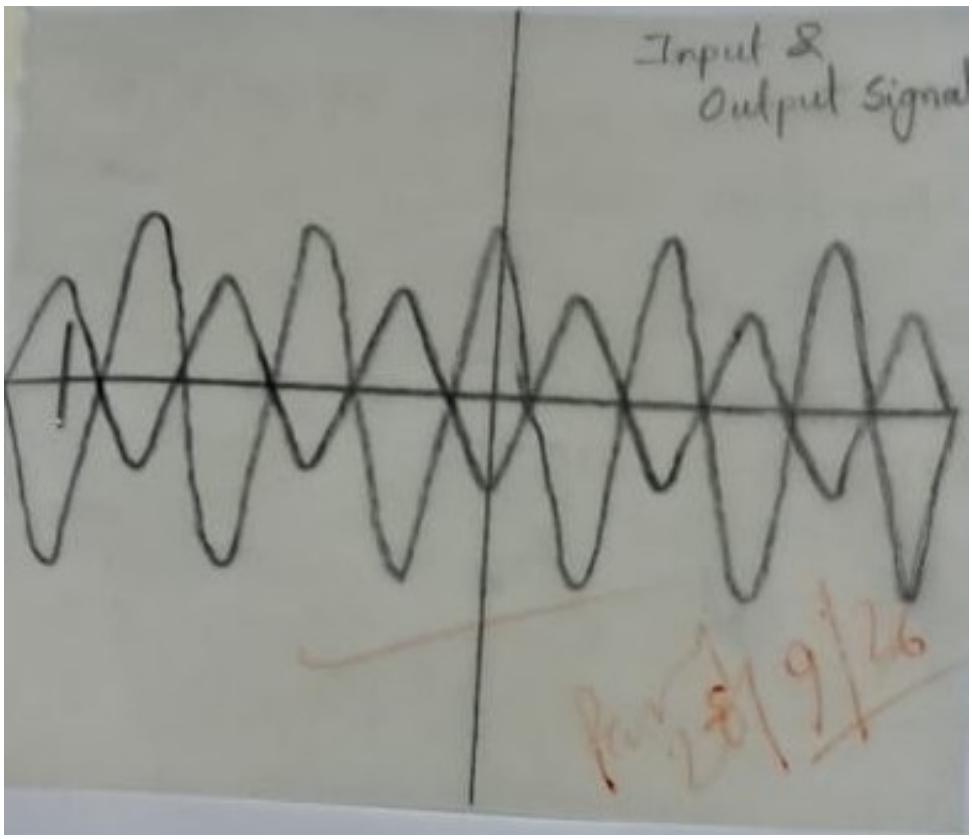


Figure 4: Oscilloscope trace record showing both input and output signal waveforms for phase comparison.

Result Analysis

The BJT Common-Emitter pre-amplifier circuit successfully established a DC bias network and amplified the small-signal AC input. Evaluation of the node voltages demonstrated reasonable convergence with theoretical calculations for the Base, Emitter, and Collector voltages, maintaining error margins beneath 10%. However, the calculated Collector Current (I_C) exhibited a significant deviation, increasing from a theoretical 1.44 mA to a measured 2.49 mA. This resulted in a substantial 72.92% error, which was independently verified to be the direct result of a component/equipment fault during the experimental setup rather than a mathematical breakdown of the voltage-divider biasing principle. The oscilloscope traces effectively visualized the necessary signal amplifications and phase alignments.

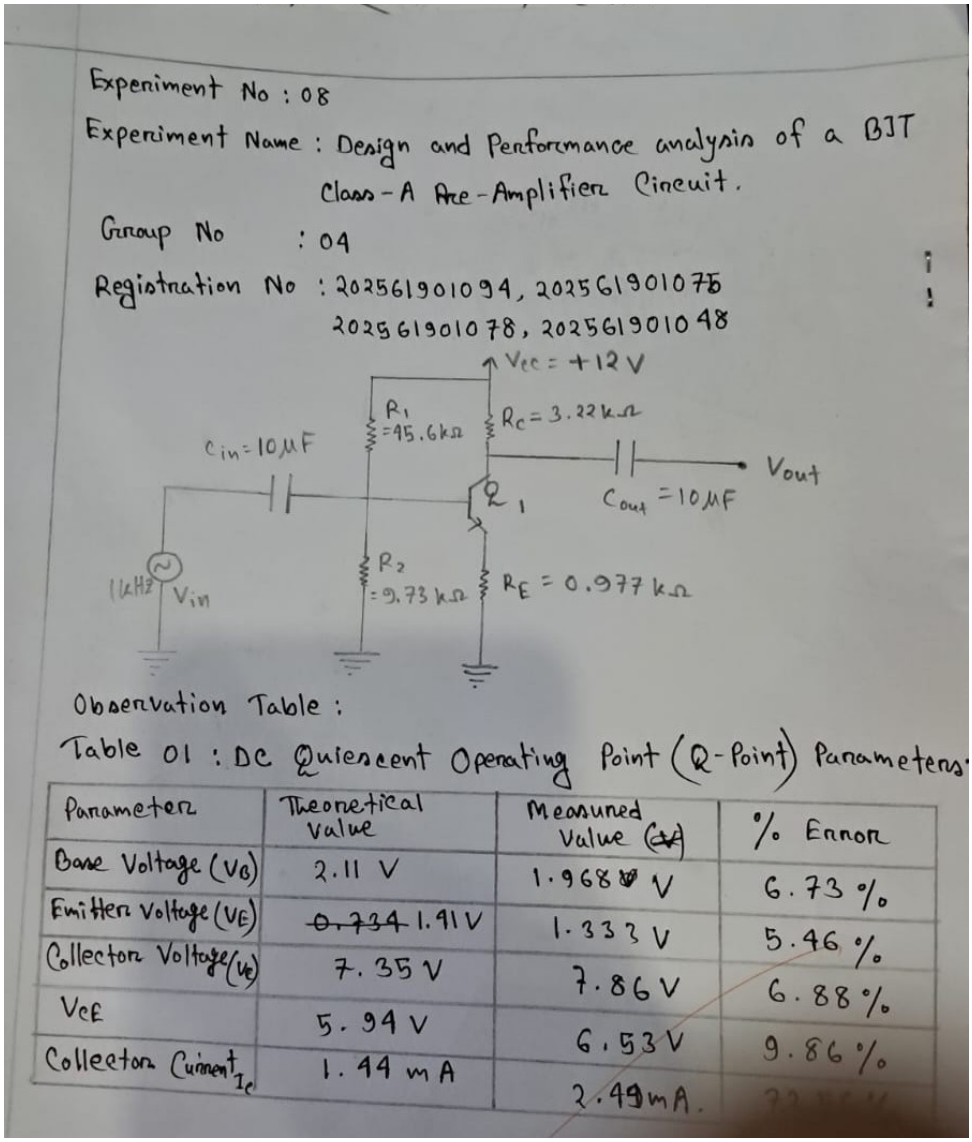
Conclusion

The Class-A Pre-Amplifier circuit was systematically built and characterized. While the overall voltage measurements validated the theoretical DC Q-point mechanics established by the bias resistors, the isolated current discrepancy highlighted the strict sensitivity of analog transistor circuits to practical hardware faults. The experiment successfully demonstrated basic BJT small-signal AC behavior, the critical role of coupling capacitors, and voltage phase inversion, proving the fundamental principles of linear signal amplification.

References

1. Boylestad, R. L., & Nashelsky, L. (2013). *Electronic Devices and Circuit Theory* (11th ed.). Pearson.
2. Alexander, C. K., & Sadiku, M. N. O. (2016). *Fundamentals of Electric Circuits* (6th ed.). McGraw-Hill Education.
3. Sedra, A. S., & Smith, K. C. (2014). *Microelectronic Circuits* (7th ed.). Oxford University Press.

Appendix: Signed Data Sheets



Experiment No : 08

Experiment Name : Design and Performance analysis of a BJT Class-A Ace-Amplifier Circuit.

Group No : 04

Registration No : 2025619010094, 2025619010075
2025619010078, 2025619010048

Observation Table :

Table 01 : DC Quiescent Operating Point (Q-Point) Parameters.

Parameter	Theoretical value	Measured Value	% Error
Base Voltage (V_B)	2.11 V	1.9688 V	6.73 %
Emitter Voltage (V_E)	~~0.734~~ 1.91 V	1.333 V	5.46 %
Collector Voltage(V_C)	7.35 V	7.86 V	6.88 %
V_{CE}	5.94 V	6.53 V	9.86 %
Collector Current I_C	1.44 mA	2.49mA.	

Figure 5: Instructor-verified experimental data collection sheet (Page 1).

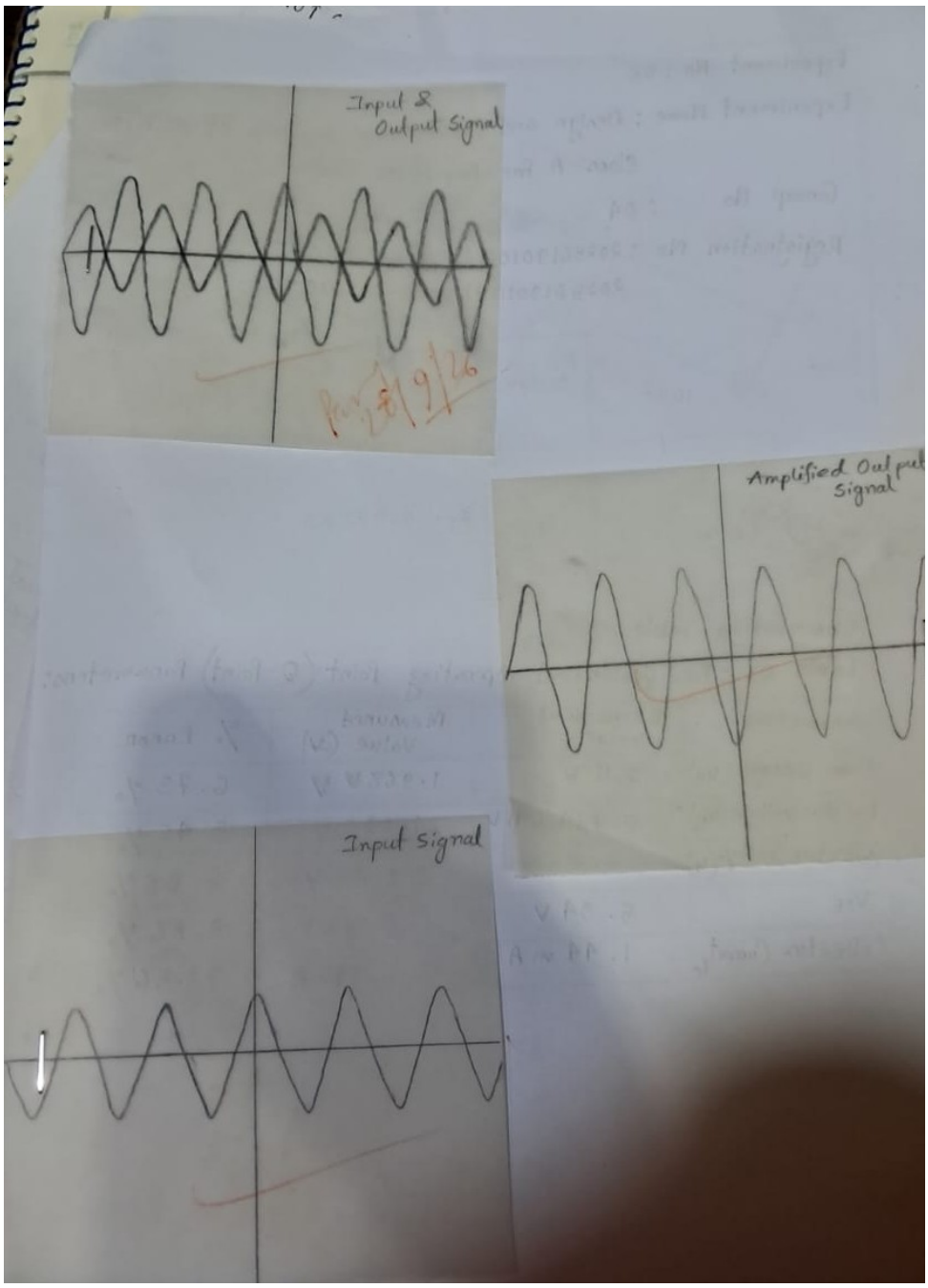


Figure 6: Instructor-verified experimental data collection sheet (Page 2).